

Summary

$$y_i = \mu + e_i$$

- 상수항만 있는 경우

$$\hat{\mu} = \frac{1}{n} \sum y_i$$

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum e_i^2$$

- 단순회귀분석

$$y_i = \beta_1 + \beta_2 x_i + e_i$$

$$\hat{\beta}_1 \quad \hat{\beta}_2$$

- 다중회귀분석

$$Y = X\beta + e$$

$$\hat{\beta} =$$

$$K \times 1$$

$$n \times 1$$

- 단항회귀

- 이중회귀

$$E(\hat{\beta}) = \beta$$

$$\text{var}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

$$\hat{\beta} \sim ?$$

* 가설검정

$$e \sim N$$

$$Y \sim N$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$\hat{\beta} \sim N$$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1})$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & \dots & \dots & a_{nk} \end{bmatrix}$$

$$\text{var}(\hat{\beta}_j) = a_{jj} \sigma^2 \quad V_j$$

계량경제학 강의 9

다중회귀모형(2)

1. 가설검정 (1)

예) $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + e_i$

$H_0: \beta_2 = 0 \text{ vs } H_1: \beta_2 > 0$

$\hat{\beta} \sim N(\beta, \sigma^2(X'X)^{-1})$

$Cov(\hat{\beta}) = \sigma^2(X'X)^{-1} = \sigma^2 \begin{bmatrix} a_{11} & a_{12} \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} \cdots & a_{kk} \end{bmatrix}$

$\hat{\beta}_2 \sim N(\beta_2, \sigma^2 a_{22})$

1) H_0, H_1

2) Decision Rule

If $\beta_2 > 0$, we reject H_0

- test statistic } ① H_0 , 불검. ✓

② " , 계산 가능 ✓

1. 가설검정(1)

- 검정 통계량

$$\hat{\beta}_2 \sim N(\beta_2, \sigma^2 a_{22})$$

- ① σ^2 이 알려져 있는 경우 : $z = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\sigma^2 a_{22}}} \sim N(0,1)$

- ② σ^2 이 알려져 있지 않은 경우 : $t = \frac{\hat{\beta}_2}{\sqrt{\hat{\sigma}^2 a_{22}}} \sim t_{(n-k)}$
k=4

If t-value $\gg 0$, we reject H_0

$\Rightarrow \beta_2$: significant.

1. 가설검정(2)

- 귀무가설 1개, 회귀계수가 2개 인 경우

예) 콥 - 더글라스 생산함수 $Y = A(K)^{\alpha}(L)^{\beta}$
 $\Rightarrow \ln Y = \ln A + \alpha \ln K + \beta \ln L + e$

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + e_i$$

가설
 $H_0: \beta_2 + \beta_3 = 1$ vs $H_1: \beta_2 + \beta_3 \neq 1$

w

$w=1$

vs.

$w \neq 1$

$\beta_2 + \beta_3 = 1$: 규준에 대한
 수치가 불변
 $\beta_2 + \beta_3 > 1$: 규준에 대한
 수치가 ↑
 $\beta_2 + \beta_3 < 1$: 수치가 ↓

검정통계량을 구하기 위해 $\hat{\beta}_2 + \hat{\beta}_3$ 의 분포를 알아야 한다.

① Under H_0 계량 - 기하
 ② " " 분포

1. 가설검정(2)

$$E(\hat{\beta}_2) = \beta_2, \quad E(\hat{\beta}_3) = \beta_3$$

- $E(\hat{\beta}_2 + \hat{\beta}_3) = \beta_2 + \beta_3$

- $Var(\hat{\beta}_2 + \hat{\beta}_3) =$

$$\rightarrow \hat{\beta}_2 + \hat{\beta}_3 \sim N(\beta_2 + \beta_3, \sigma^2(a_{22} + a_{33} + 2a_{23}))$$

$$Var(A + B)$$

$$= Var(A) + Var(B) + 2Cov(A, B)$$

비공분산

• 검정 통계량

- σ^2 이 알려져 있는 경우 :

$$\frac{(\hat{\beta}_2 + \hat{\beta}_3) - (\beta_2 + \beta_3)}{\sqrt{(a_{22} + a_{33} + 2 \cdot a_{23}) \sigma^2}} \sim N(0, 1)$$

- σ^2 이 알려져 있지 않은 경우 :

$$\hat{\sigma}^2$$

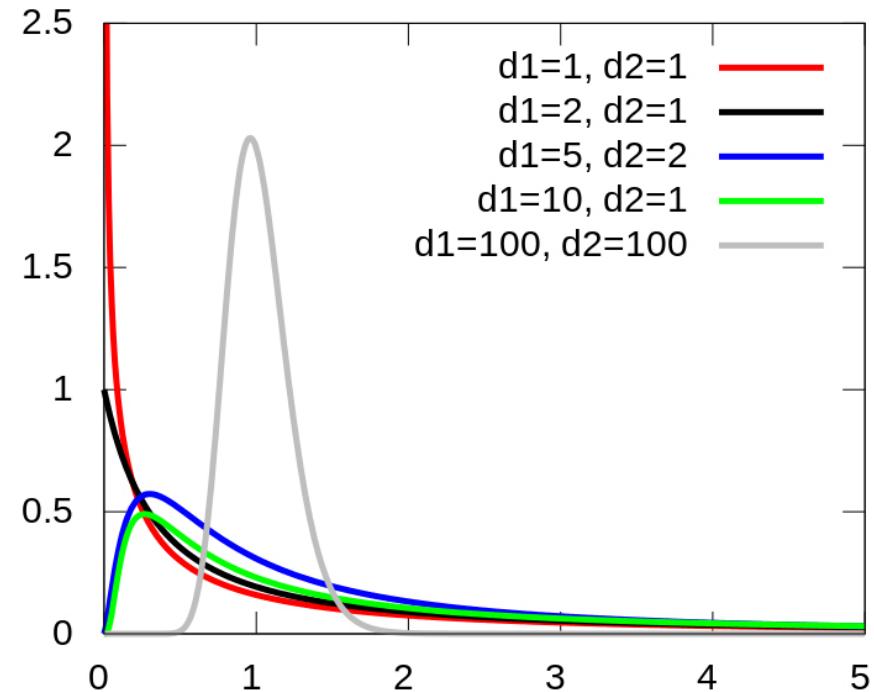
2. 복합가설 검정

Note) F-분포 : 서로 독립인 두개의 χ^2 분포의 적절한 비율

$$A \sim \chi^2(n_1)$$

$$B \sim \chi^2(n_2)$$

$$f - val = \frac{A/n_1}{B/n_2} \sim F(n_1, n_2)$$



2. 복합가설 검정

예) $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + e_i$

$H_0: \beta_2 \overset{\beta_2=0 \text{ and } \beta_3=0}{=} \beta_3 \overset{\beta_2=0 \text{ and } \beta_3=0}{=} 0$ vs $H_1: \text{적어도 하나는 0 이 아니다.}$

- Intuition.

• unrestricted regression : $Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \hat{\beta}_4 X_{4i} + \hat{e}_i$
 $\Rightarrow \Sigma \hat{e}_i^2$

✓ restricted regression : $Y_i = \hat{\beta}_1^* + \hat{\beta}_4^* X_{4i} + \hat{e}_i$

If H_0 is true,
 " wrong

$\Rightarrow \Sigma \hat{e}_i^{*2}$
 $\boxed{\Sigma \hat{e}_i^2 \approx \Sigma \hat{e}_i^{*2}}$
 $\Sigma \hat{e}_i^2 \ll \Sigma \hat{e}_i^{*2}$

2. 복합가설 검정

만일 귀무가설이 사실이라면 $\sum \hat{e}_i^2 \approx \sum \hat{e}_i^{*2}$ 일 것이다.

만일 귀무가설이 사실이 아니라면 $\sum \hat{e}_i^2 \ll \sum \hat{e}_i^{*2}$

Restricted $\frac{\sum \hat{e}_i^{*2}}{\sigma^2} \sim \chi^2(n-2)$ $\nwarrow$ regressor 2

Unrestricted $\frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-4)$

$\left. \begin{array}{l} \frac{\sum \hat{e}_i^{*2}}{\sigma^2} \sim \chi^2(n-2) \\ \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-4) \end{array} \right\} \frac{\sum \hat{e}_i^{*2}}{\sigma^2} - \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(2)$

2. 복합가설 검정

= f-value $\checkmark \chi^2$

$$\frac{\left(\frac{\sum \hat{e}_i^{*2}}{\sigma^2} - \frac{\sum \hat{e}_i^2}{\sigma^2} \right) / 2}{\frac{\sum \hat{e}_i^2}{\sigma^2} / (n-4)} \sim F(2, n-4)$$

• 일반적으로 $\checkmark \chi^2$

- j: 귀무가설에서 제약조건의 개수
- k: 설명변수의 개수

← χ^2 가설 검정

$$\text{f-value} = \frac{(\sum \hat{e}_i^{*2} - \sum \hat{e}_i^2) / j}{\sum \hat{e}_i^2 / (n-k)} \sim F(j, n-k)$$

* $H_0: \beta_2 = 0$ and $\beta_3 = 0$

$\checkmark$ t-value₁ = $\frac{\hat{\beta}_2 - 0}{\sqrt{\text{var}(\hat{\beta}_2)}} \sim t(n-k) \Rightarrow H_0: \beta_2 = 0$
 $\beta_3 \neq 0$

$\checkmark$ t-value₂ = $\frac{\hat{\beta}_3 - 0}{\sqrt{\text{var}(\hat{\beta}_3)}} \sim t(n-k) \Rightarrow H_0: \beta_3 = 0$
 $\beta_2 \neq 0$

(A) Simple hypothesis testing

$$H_0: \beta_1 = 1$$

$$H_0: \beta_1 = 0$$

$$H_0: \beta_1 + \beta_2 = 0$$

$$H_1: \beta_1 \neq 1$$

$$H_1: \begin{matrix} < 0 \\ > 0 \end{matrix}$$

(B) Joint hypothesis testing

$$H_0: \beta_1 = 0 \text{ and } \beta_2 = 0$$

$$H_1: \beta_1 \neq 0 \text{ or } \beta_2 \neq 0, \text{ not both } = 0.$$

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \epsilon_i$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $\hat{\beta}_1 \quad \hat{\beta}_2 \quad \hat{\beta}_3$

✗ Q1) $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + e_i$

$$i = 1 \dots n$$

$$k = 4$$

II

$H_0: \beta_2 = 0 \text{ and } \beta_3 = 0$

$J = 2$

(# of restriction)

$H: \text{Both } \beta_2 \text{ and } \beta_3 \neq 0$

"F-test"

2 If $H_0: \text{true}$, $\frac{\sum \hat{e}_i^2}{\sum e_i^2}$

If $H_0: \text{not correct}$, $\frac{\sum \hat{e}_i^2}{\sum \hat{e}_i^{*2}}$

$\Rightarrow$ A general approach.

Unrestricted Reg.	Restricted Reg.
$\textcircled{1} Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \hat{\beta}_4 X_{4i} + \hat{e}_i$ $\Rightarrow \sum \hat{e}_i^2$	$\textcircled{2} Y_i = \hat{\beta}_1^* + \hat{\beta}_4^* X_{4i} + \hat{e}_i^*$ $\Rightarrow \sum \hat{e}_i^{*2}$

$$[3] \text{ Q1) } \sum \hat{e}_i^{*2} - \sum \hat{e}_i^2$$

$\Rightarrow$ 타진

① $H_0: F$, $\mu_1 \neq \mu_2$ ok

~~②~~ " $\mu_1 = \mu_2$ known

$$[4] \frac{\sum \hat{e}_i^{*2} - \sum \hat{e}_i^2 / J}{\sum \hat{e}_i^2 / (n-k)} = F\text{-value} \sim F(J, n-k)$$

<예> ① Unrestricted, $e_i \sim iid N(0, \sigma^2)$

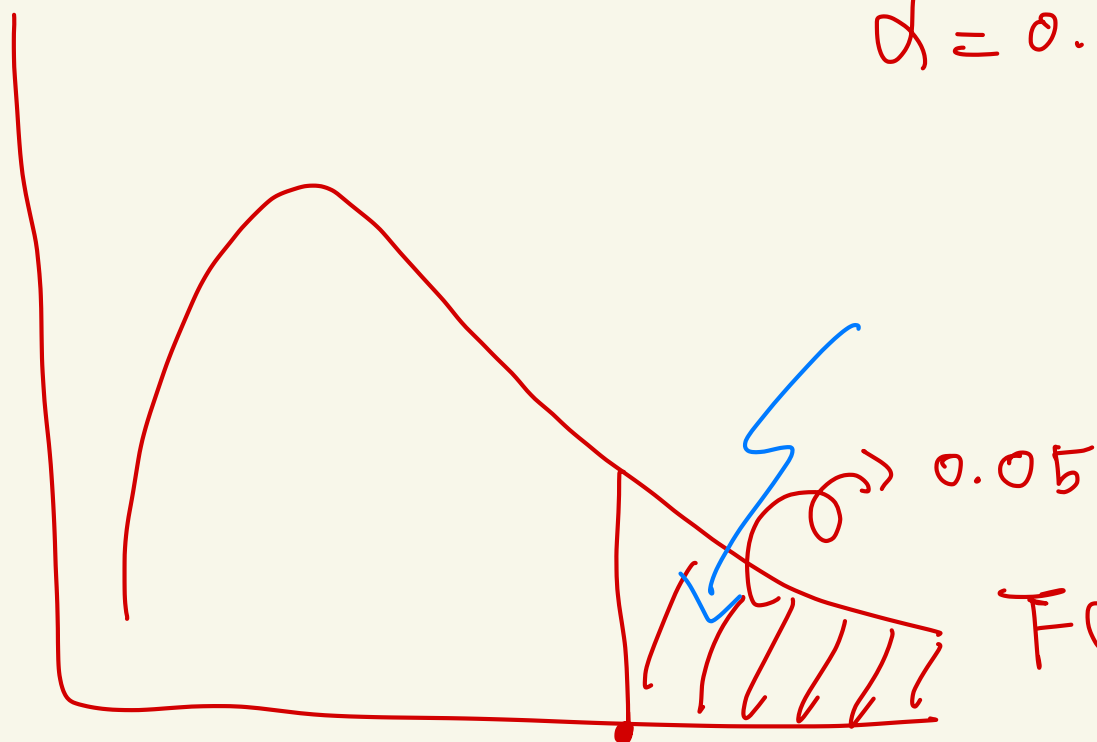
② Restricted

① $\frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(\underline{\underline{n-4}})$

② $\frac{\sum \hat{e}_i^{*2}}{\sigma^2} \sim \chi^2(\underline{\underline{n-2}})$

$$\frac{(2) - (1)}{(n-4) - (n-2)} = \frac{\frac{\sum \hat{e}_i^{*2} - \sum \hat{e}_i^2}{\sigma^2} / 2}{\frac{\sum \hat{e}_i^2}{\sigma^2} / (n-4)} \sim F(2, 4)$$

$$\alpha = 0.05$$



$$\textcircled{1} \underline{F\text{-value} > \star},$$

to reject

$$\textcircled{2} p\text{-value} < \alpha.$$

we reject H_0

Note

A n v U $Z \sim N(0, 1)$

$Z^2 \sim \chi^2(1)$

Optimization

$n \rightarrow \infty$

$t\text{-value} \sim t_{\frac{\alpha}{2}, n-1}$

$(t\text{-value})^2 \sim F_{\frac{\alpha}{2}, n-1}$

$= F\text{-value}$

$n \rightarrow \infty$